

# ECE 251C – Fall 2025 – Parhi

## Quiz 1

**Quiz Rules:** no notes, computers, calculators, or phones; have only a pen/pencil; work independently

Please show your work and provide any justification of your answers so that you can receive partial credit.

*On my honor, I pledge that I have neither given nor received unpermitted aid on this exam.*

**signature:**

**name:**

(a) \_\_\_\_\_ / 5 points

(b) \_\_\_\_\_ / 4 points

(c) \_\_\_\_\_ / 4 points

(d) \_\_\_\_\_ / 5 points

(e) \_\_\_\_\_ / 7 points

**Total:** \_\_\_\_\_ / 25 points

Consider a discrete-time filter with real coefficients whose frequency response is

$$H(e^{j\omega}) = (1 - 2\alpha \cos \omega + \alpha^2)e^{-j\omega}, \quad 0 < \alpha < 1.$$

- (a) Determine the corresponding **impulse response**  $h[n]$  and clearly state its **length** and **support** (the range of  $n$  for which  $h[n] \neq 0$ ).

**Solution.** Recall that  $2 \cos \omega = e^{j\omega} + e^{-j\omega}$ . Therefore,

$$H(e^{j\omega}) = (1 - \alpha e^{j\omega} - \alpha e^{-j\omega} + \alpha^2)e^{-j\omega}. \quad (1)$$

Thus,

$$H(z) = (1 - \alpha z - \alpha z^{-1} + \alpha^2)z^{-1} = z^{-1} - \alpha - \alpha z^{-2} + \alpha^2 z^{-1} = -\alpha + (1 + \alpha^2)z^{-1} - \alpha z^{-2}. \quad (2)$$

In which case, we have that  $\text{supp}(h) = \{0, 1, 2\}$ , the length is  $|\text{supp}(h)| = 3$ , and  $h[0] = -\alpha$ ,  $h[1] = (1 + \alpha^2)$ , and  $h[2] = -\alpha$ .

- 3 points: correct filter coefficients with clear derivation from frequency response to coefficients
- 1 point: correct support
- 1 point: correct length

- (b) Determine whether this filter has **linear phase**, and if so, specify its **phase response**.

**Solution.** From (a) we see that the filter has an odd number of taps and is symmetric so it is Type-I linear-phase. From the form of  $H(e^{j\omega})$ , we already see that it is written in the amplitude-phase decomposition so we immediately see that the phase response is  $\phi(\omega) = -\omega$ .

- 2 points: correct argument (with justification) for the filter being linear phase
- 2 points: correct phase response (with justification)

- (c) Is the filter **causal**? If not, explain how to construct a causal version.

**Solution.**  $h[n]$  is nonzero only for  $n = 0, 1, 2$  so it is causal.

- 2 points: correct statement about causality
- 2 points: correct argument (with justification) for causality by mentioning the support at  $n \geq 0$ .

- (d) Suppose we define

$$\tilde{H}(z) = H(z)H(z^{-1}).$$

Is  $\tilde{H}(z)$  guaranteed to be **linear phase**? Justify your answer.

**Solution.** We have that

$$\tilde{H}(e^{j\omega}) = H(e^{j\omega})H(e^{-j\omega}) = |H(e^{j\omega})|^2 \quad (3)$$

which has linear (zero) phase.

- 2 points: correct conclusion of zero phase

- 3 points: correct argument (with justification) for zero phase. It is correct to have done it abstractly as above or with the specific frequency response defined at the beginning of the question.

(e) Define

$$G(z) = H(z) + H(z^{-1}).$$

(i) Express  $G(e^{j\omega})$  without computing  $g[n]$ .

**Solution.** From the solution to (a) we have that

$$H(z) = -\alpha + (1 + \alpha^2)z^{-1} - \alpha z^{-2} \quad (4)$$

and so

$$H(z^{-1}) = -\alpha + (1 + \alpha^2)z - \alpha z^2 = z^2 H(z). \quad (5)$$

Therefore,

$$G(e^{j\omega}) = H(e^{j\omega}) + e^{j2\omega} H(e^{j\omega}) \quad (6)$$

$$= (1 + e^{j2\omega}) H(e^{j\omega}) \quad (7)$$

$$= (1 + e^{j2\omega})(1 - 2\alpha \cos \omega + \alpha^2) e^{-j\omega} \quad (8)$$

$$= 2 \cos \omega (1 - 2\alpha \cos \omega + \alpha^2). \quad (9)$$

- 3 points: correct answer with justification (without going through  $g[n]$ ). All or nothing.

(ii) Determine whether  $G(z)$  is **linear phase**, and if so, specify the **phase response** and **type**.

**Solution.** Since  $G(e^{j\omega})$  is purely real, it has linear (zero) phase, i.e.,  $\phi(\omega) = 0$ . To quickly see the type, observe that it is an even function of  $\omega$  and thus it must be symmetric about  $n = 0$  which implies that it must be Type-I.

- 1 point: correct conclusion of zero phase with justification
- 1 point: correct type (Type-I) with justification

(iii) Find  $g[n]$ , its **support**, whether it is **causal**.

**Solution.** In the time domain,

$$g[n] = h[n] + h[-n]. \quad (10)$$

Therefore,

$$g[n] = \begin{cases} -\alpha, & \text{if } n = -2, \\ 1 + \alpha^2, & \text{if } n = -1 \\ -2\alpha, & \text{if } n = 0 \\ 1 + \alpha^2, & \text{if } n = 1 \\ -\alpha, & \text{if } n = 2, \\ 0, & \text{else.} \end{cases} \quad (11)$$

Thus,  $\text{supp}(g) = \{-2, -1, 0, 1, 2\}$ . Since  $g[n] \neq 0$  for  $n = -2$  and  $n = -1$ , it is not causal, but it could be made causal by delaying by 2, i.e.,  $g_{\text{causal}}[n] = g[n - 2]$ .

- 1 point: correct coefficients and support
- 1 point: correct statement about causality (with justification)